How to use NanoServe

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After SETUP.md or `./install.sh`, activate the environment:

```
source .venv/bin/activate && source .env.nanoserve
```

Start the server

Open **http://localhost:8000** for the Web UI.

Web UI

1. Enter a prompt in the text box.
2. Choose **Compute Engine**: CPU, GPU, or Auto.
3. Click **Generate** - response shows tokens and which device ran.

HTTP API

```
curl -s http://localhost:8000/health   jq .
curl -X POST http://localhost:8000/v1/completions \
  -H 'Content-Type: application/json' \
  -d '{"prompt":"Explain quantum dots in one sentence.","max_tokens":48,"device":"auto"}'
```

Fields:
- `prompt` - input text
- `max_tokens` - max generated tokens (default 64)
- `device` - `cpu`, `gpu`, or `auto` (GPU falls back to CPU with warnings)

Python SDK

```
pip install -e .
python examples/sdk_demo.py
```

```
from nanoserve import NanoServe
engine = NanoServe(device="auto")
text = engine.generate("Hello from the SDK", max_tokens=32)
print(text)
```

Async:

```
import asyncio
from nanoserve import NanoServe
async def main():
    engine = NanoServe(device="cpu")
    text = await engine.generate_async("Async hello", max_tokens=24)
    print(text)
asyncio.run(main())
```

Terminal UI (TUI)

```
pip install httpx rich
python tui/client.py --url http://localhost:8000 --device auto
```

Slash commands in the TUI:
- `/device cpu gpu auto` - switch backend
- `/quit` - exit

Load testing

50 /

```
python3 tests/load_test_report.py --preset 50
python3 tests/load_test_report.py --preset 150
python3 tests/load_test_report.py --preset 300
```

Device selection summary

Troubleshooting

Reports: reports/FULL_TEST_REPORT.md, [reports/STRESS_REPORT.md](reports/STRESS_REPORT.md